



Growth during the first week of life and physiological body weight oscillation between feedings using high-frequency weighing in individually housed calves fed unrestricted amounts of milk twice daily

Anton Maria Vorndran,¹ Heinz Bernhardt,^{2,3} and Julia Steinhoff-Wagner^{1,3*}

¹Professorship of Animal Nutrition and Metabolism, TUM School of Life Sciences, Technical University of Munich, 85354, Freising, Bavaria, Germany

²Chair of Agricultural Systems Engineering, TUM School of Life Sciences, Technical University of Munich, 85354, Freising, Bavaria, Germany

³HEF World Agricultural Systems Centre, Technical University of Munich, 85354, Freising, Bavaria, Germany

ABSTRACT

This study investigated early postnatal weight dynamics in 19 Brown Swiss calves (11 male, 8 female) during their first week of life. It was hypothesized that high-frequency weight monitoring during the first week of life in neonatal calves could reveal patterns of postprandial weight loss and early growth dynamics, which may be used to estimate metabolic activity. Calves were housed individually in hutches on calibrated scales that recorded BW every 10 s, enabling high-resolution monitoring of growth and postprandial weight changes. All calves were fed their dam's milk twice daily in unrestricted amounts for 14 feedings. Urine was quantitatively collected and weighed to estimate the proportion of weight loss attributable to metabolism and evapotranspiration. Data were analyzed using generalized linear mixed models with sex, feeding number, hours postprandial, and their interaction as fixed effects, and calf ID as a random intercept. A first-order autoregressive correlation structure accounted for autocorrelation in repeated measurements. Based on significant differences in weight-loss dynamics, feedings were grouped into early (1–3) and later (4–14) feedings. Sawtooth-like growth curves were observed for weight development during the first week of life due to daily fluctuations in BW with maxima occurring after feeding and minima before the next feeding. The calves showed an ADG of 1.31 ± 0.28 kg during the first week of life. Postprandial weight loss ranged from 3.5% to 7.4% of BW over 10 to 14 h and was significantly influenced by the feeding event and the duration of the postprandial period. The rate of weight loss was lowest during the first 3 feedings (0.346% BW/h), increasing and stabilizing at 0.607% BW/h at later feedings. Fourth-

degree polynomial models best described the hourly postprandial weight loss patterns for feedings 1 to 3 and 4 to 14 ($R^2 = 0.994$ and 0.999 , respectively). The highest weight loss occurred in the first hour postprandial and weight loss continuously decreased thereafter. Metabolization and evapotranspiration losses, calculated by the separation of the collected urine amounts from total weight loss observed on the scales, accounted for 79% of total weight loss at the first feeding and 33% to 49% in subsequent feedings. Strong positive correlations were found between postprandial weight loss to urine excretion ($r = 0.729$), and milk intake ($r = 0.811$). The findings highlight the dynamic nature of BW and the value of high-resolution, continuous monitoring (1 measurement every 10 s) in early calf development.

Key words: colostrum, scale, shrink, postnatal, calf

INTRODUCTION

Accurate weighing of animals is a fundamental aspect of scientific research, playing a crucial role in various fields such as animal nutrition, veterinary medicine, and zoology. Precise weight measurements provide the basis for assessing animal health, growth, and nutritional status. The gold standard of direct weight measurement is using a calibrated electronic scale to determine the BW of a calf. Indirect measurements include using a heart girth tape, a partial scale, a hipometer, or image analysis with artificial intelligence (Dingwell et al., 2006; Cantor et al., 2020; Bai et al., 2025). Partial scales (e.g., forefront scales) have been reported to yield satisfactory results when attached to an automated milk feeder (Cantor et al., 2020). Additionally, image analysis, especially with advancements in artificial intelligence, has achieved high accordance with direct weight measurements of 97% (Bai et al., 2025). Other indirect techniques, such as the heart girth tape or the hipometer, provide reasonably accurate estimates but are more susceptible to

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*Corresponding author: jsw@tum.de

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

measurement error due to user-dependent factors, such as the amount of pressure applied or the positioning of the tool on the animal (Dingwell et al., 2006). All these techniques are single measurements conducted at regular intervals, based on which the desired parameter BW gain can be calculated for a given period. The first week of life in neonatal calves is characterized by profound physiological changes, including passive immunization through colostrum consumption (Godden et al., 2019), proteinuria (Pierce, 1959), rapid proliferation and growth of the gastrointestinal epithelium (Blum and Baumrucker, 2002; Steinhoff-Wagner et al., 2014), and the establishment of intestinal microbial populations (Fischer et al., 2018). The success of these adaptation processes is critical for survival (Donovan et al., 1998; Godden et al., 2019). During this period, an animal's body weight is highly dynamic, prone to increase through feed and water consumption and decrease through evapotranspiration, urination, defecation, and metabolization (Coffey et al., 2001; Vorndran et al., 2023). Among the external factors influencing these dynamics, housing conditions can affect physical activity, energy expenditure, and nutrient intake (Donadio et al., 2025). In particular, group housing has been shown to promote greater social interaction and activity. Socially housed calves are typically more active and consume more feed during the preweaning and postweaning phases, leading to higher energy turnover and improved weight gain compared with individually housed calves (Costa et al., 2016; Donadio et al., 2025). The postprandial decrease in BW, often referred to as shrink (Coffey et al., 2001), represents a physiological response following feed intake in neonates and adult animals. It encompasses processes such as the elimination of excreta and urine, shifts in water balance, and respiration, all of which contribute to transient reductions in BW after feeding (Self and Gay, 1972; Phillips et al., 1991; Coffey et al., 2001). In neonatal calves, these fluctuations are particularly pronounced because marked metabolic and endocrine changes occur during the immediate neonatal period influencing immunity, glucose metabolism, the somatotrophic axis and gastrointestinal development (Hammon and Blum, 1997; Blum and Hammon, 1999; Bittrich et al., 2002; Steinhoff-Wagner et al., 2014). As such, postprandial weight loss could serve as an indicator for metabolic adaptation and homeostatic efficiency during early life, potentially offering valuable insight into the animal's physiological development and overall health status. In the absence of solid feed intake, high amounts of milk consumption with a low DM concentration lead to a fluid intake that is large in relation to the calf's small BW. As a result, urine accounts for a considerable proportion of postprandial weight loss, which also reflects the calf's fluid balance and health status, especially in cases of dehydration or neonatal

diarrhea. Therefore, evaluating this process is essential to accurately distinguish short-term physiological weight changes from actual growth, particularly when assessing ADG during the preweaning period. Continuous weight monitoring at small measurement intervals offers valuable insights into these dynamics. It detects subtle changes that might be missed with less frequent measurements, revealing patterns related to feeding, urine excretion, or animal activity. This level of detail is particularly valuable in studies aimed at optimizing feeding strategies, improving animal welfare, and enhancing productivity in livestock management. The aim of this study was to investigate how close weight monitoring during the first week of life could reveal patterns in weight development and postprandial weight loss in calves. High-frequency weight measurements were used to describe early growth dynamics and short-term weight fluctuations. Furthermore, the study provided a first attempt of exploring the possibility of differentiating weight loss attributable to urination from that due to metabolization and evapotranspiration losses.

MATERIALS AND METHODS

Experimental Design

Nineteen full-term Brown Swiss calves ($287.12 \pm [SE] 1.34$ d), 11 males and 8 females, with a mean birth weight of 44.63 ± 1.29 kg were housed in commercially available calf hutches (length 164 × width 100 × height 107 cm, no. 14611, 172 kg, Albert Kerbl GmbH, Buchbach, Germany) placed on certified animal scales (As Wägetechnik GmbH, Garching-Hochbrück, Germany) inside the experimental facility under controlled environmental conditions (mean temperature $17.34^{\circ}\text{C} \pm 0.84^{\circ}\text{C}$, mean relative humidity $64.09\% \pm 1.25\%$, KlimaLoggPro, TFA Dostmann, Germany). Only calves that required no more than minimal assistance at birth were included in this study. Minimal assistance was defined as support without the use of tools such as ropes or other equipment. Initially, 21 calves were enrolled in the study; however, 2 calves were excluded from the dataset, 1 due to illness and 1 due to incomplete urine collection by human error during the trial. The scales built by AS Wägetechnik GmbH used a BWBI 2015 1000HR Fli weighing cell with a digital weighing terminal Type AS-WT-715. The used scales were calibrated by AS Wägetechnik GmbH, which are accredited (Certificate D-K-17702-01-00) by the Deutsche Akkreditierungsstelle. The scales used in this study were calibrated for a maximum capacity of 1,000 kg and had a measurement sensitivity of ± 0.1 kg. All scales were initially calibrated and tested for continuous accuracy using a certified 50-kg calibration weight (Zwiebel certificate number: Z22 38274). The weight was

placed on each of the 4 scales, which were then turned on and left undisturbed for 1 wk to assess whether the scales could consistently document the weight of the calves with high accuracy throughout the experiment. The scales recorded the calves' weight every 10 s, together with a timestamp within the stated measurement sensitivity of 0.1 kg. The facility was equipped with 4 stationary scales and 5 mobile calf hutches that could be used simultaneously. Hutch and scale assignments were conducted by farm personnel using a randomized approach, contingent upon hutch availability following routine cleaning, drying, and disinfection procedures. Calves were enrolled in the trial sequentially and allocated to available hutches accordingly. To ensure balanced utilization of the weighing scales and minimize potential systematic bias, each newly enrolled calf was assigned to a different hutch and scale combination than its predecessor. All animals were housed on straw bedding as mandated by German animal welfare law (§§1–11; Tierschutz-Nutztierhaltungsverordnung, 2021) for the first 7 d of life. The straw was weighed before provision (1.25 ± 0.16 kg/calf and week). Each hutch was fitted with a urine tray beneath the slatted flooring, allowing excreted urine to leave the scale and be collected in a canister. Feces was retained by the bedding material and was collected, weighed, and dried at the end of the 7-d rearing period. The experiment was terminated before the 15th feeding for each calf at the usual feeding time and all calves were cross-weighed on a second scale to confirm the weight measurements of the scale on which they were initially placed. All routine procedures and human-calf interactions were documented in individual color-coded folders using standardized record sheets. Human interference during weighing was verified by identifying abrupt increases in recorded weights (e.g., calf weight plus a person's BW), which consistently aligned with documented interventions or feeding times.

Feeding Schedule

All animals were fed twice daily at 0700 h and 1700 h with an unrestricted volume of warm dam's milk ($>30^{\circ}\text{C}$), measured by the weight difference of consumed to unconsumed milk residues from the teat bucket. Initially, all calves were separated from their dams immediately after birth, transferred to the experimental barn and received colostrum on average 82.31 ± 7.69 min (within 42–136 min) after birth. All calves received colostrum, transition milk, and whole milk from their individual dam and no milk pool was used in the trial. The quality of the supplied colostrum was measured using a digital Brix refractometer (Milwaukee Instruments MA871, Szeged, Hungary) with an average colostrum quality supplied to the calves of 22.72° Brix $\pm 0.75^{\circ}$ Brix. Calves were permitted to drink from the teat bottle

or teat bucket for as long as they desired, and feeding was discontinued only when they no longer showed interest in drinking or lay down independently, disregarding the bottle or bucket. The ingested milk amounts are therefore to be interpreted as fed until satiety under twice daily feeding conditions. The calves received no additional feed in the form of hay, calf starter, or water during their initial week of life. In total, the animals were fed 14 times during the first week of life.

Analytics

Laboratory analysis was conducted for samples of colostrum, milk and urine. Upon arrival at the laboratory, each sample was assigned an identification number, which prevented a link to the corresponding animal by the analysts. The methods used for analysis have been published in detail in earlier works by the authors (Vorndran and Steinhoff-Wagner, 2025; Vorndran et al., 2025). Briefly, dams' milk was analyzed according to the official methods given by the VDLUFA (2020) for DM, CP, crude ash, and fat. Gross energy was determined by calorimetry of freeze-dried milk samples using a bomb calorimeter (C2000 basic with a KV 500 digital cooler; IKA-Werke GmbH & Co. KG, Staufen, Germany). Lactose was analyzed using a photometric detection kit (Enzytec Lactose/ D-Galactose, R-Biopharm AG, Darmstadt, Germany). Dry matter and crude ash were determined by weight difference after drying at 103°C overnight and after burning the sample at 550°C for 6 h. Crude protein was measured by nitrogen analysis with the Kjeldahl method. Fat was analyzed using the method of Weibull Stoldt. Analogue, to the milk analytics, urine was analyzed for DM, nitrogen, and crude ash. Urea was analyzed in urine using photometric detection following the use of an enzyme kit (Enzytec Liquid Urea, R-Biopharm AG, Darmstadt, Germany).

Data Processing and Analysis

The scales recorded data in a text file, which was subsequently imported into Microsoft Excel (2016; Microsoft Corp.) for analysis. The dataset for each calf consisted of approximately 60,000 individual weight measurements. Calf weight outliers (>5 kg), only detected during feeding times, when the feeding bucket was attached to the calf hutch, or during periods in which staff interacted with the calves for health assessment in a way that disrupted the scale readings were manually removed from the dataset. The number of removed data points was individual for each calf but accounted for approximately 2,500 weight measurements per calf (5% of all data points). Following outlier removal, the data were smoothed using a moving median filter: For each data point, the median of the

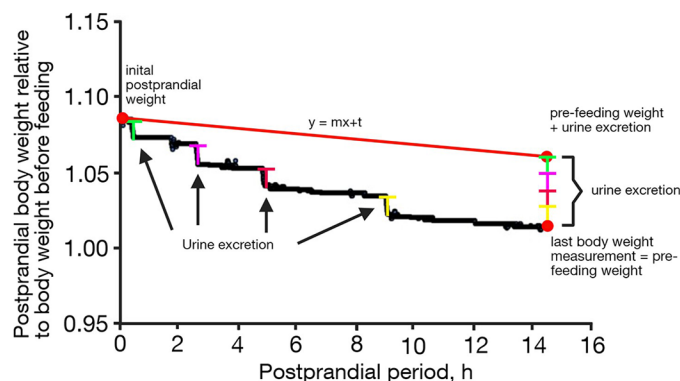


Figure 1. Illustration of the calculation of metabolism and evapotranspiration losses from postprandial weight loss data relative to prefeeding BW of neonatal Brown Swiss calves. Created in BioRender. Vorndran, A. (2025), <https://biorender.com/iyz0jb6>.

10 neighboring measurements was calculated. This smoothing technique reduced short-term fluctuations in calf BW caused by animal movement without altering the total number of data points or weight dynamics. To correct for systematic overestimation of weight due to water and feces retention in bedding material, an error adjustment was applied to the smoothed data by interpolation of the observed overestimation at the end of the experiment to each 10-s interval using a linear approach, which distributed the observed error evenly across all measurements without changing the weight dynamics. For intercalf comparisons over time, absolute BW values (kg) were normalized to a relative scale by dividing each measurement by either the last prefeeding weight or the birthweight, depending on the analysis context. For postprandial weight loss analysis, the prefeeding weight, derived from the linearly interpolated data, served as the normalization denominator. For determination of growth trends on a week basis, the birthweight was used as denominator.

All weight measurements were initially visualized, and individual weight trajectories were characterized using linear regression analyses in Microsoft Excel (2016) at 10-s intervals throughout the first week of life. For each calf, 2 regression approaches were used: one with the intercept fixed at the measured birth weight ($t = 1.0$), and one where the intercept was estimated as part of the linear regression model. These analyses were conducted solely for descriptive purposes and did not involve statistical inference. To evaluate postprandial weight loss, all feeding episodes were individually extracted from the dataset and plotted in Excel. Due to the visible polynomial trend in the data, fourth-degree polynomial regressions were then fitted to capture the dynamics of weight loss following each feeding. The processed data from the curve analytics were then analyzed in

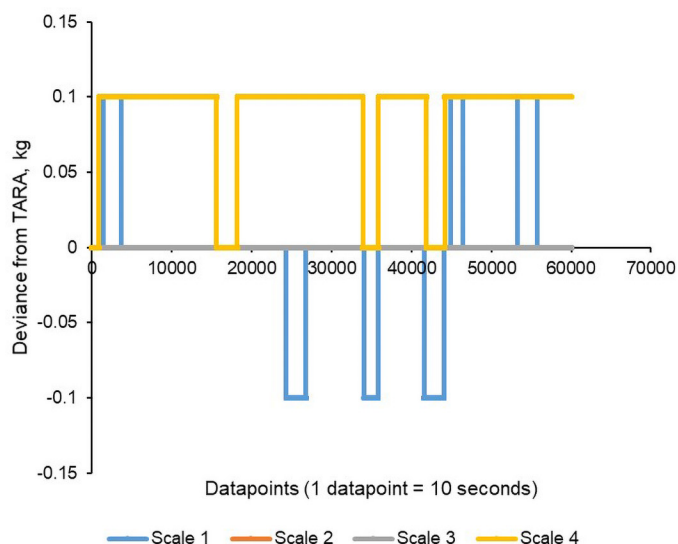


Figure 2. Verification of the used scales' sensitivity (± 0.1 kg) under repeated measurement conditions during one week under experimental conditions using a certified 50-kg calibration weight. Scales 2 and 3 showed no deviation from 0. TARA refers to resetting the display to 0 while a scale was loaded.

RStudio (version 4.4.2, Posit). Before modeling, the data were assessed for normality and homoscedasticity. Due to violations of these assumptions, a generalized linear mixed model (GLMM) was fitted using the glmmTMB package in RStudio. Spearman rank correlations were calculated in SigmaPlot 15.0 (SyStat Software GmbH, Frankfurt am Main, Germany). For 83 out of 2,660 hourly weight loss intervals, the data could not be used for modeling purposes as a result of weight loss being below the measurement sensitivity of 0.1 kg of the scales. Because log-transformation required strictly positive values, the variable postprandial weight loss was multiplied by -1 . To visualize and process postprandial weight loss data on an hourly basis, the dataset was restricted to the first 10 h after feeding, thereby excluding confounding effects from the unequal durations of postprandial periods during the day (10 h) and night (14 h). Given the continuous, positive, and right-skewed distribution of postprandial weight loss, a Tweedie distribution with a log link was used for modeling. The model included fixed effects for sex (male, female), feeding number (1–14), and hours postprandial (1–10), as well as the interaction of feeding number and hours postprandial. A natural spline with 3 df was applied to the time variable to capture the nonlinear relationship between time and weight loss. Calf ID was included as a random intercept to account for individual variability in the calves' responses over time. To account for temporal autocorrelation of repeated measures within calves, a first order autoregressive correlation structure was in-

Table 1. Composition and consumption of colostrum, transition milk, and whole milk from Brown Swiss calves during the first week of life at twice daily feedings (14 feedings) and voluntary milk intake¹

Feeding	Feed consumption, kg		DM, %		Fat, %		CP, %		Energy, MJ GE/kg		Lactose, %	
	Mean	SEM	Mean	SEM	Mean	SEM	Mean	SEM	Mean	SEM	Mean	SEM
1	2.58	0.24	24.38	0.76	7.01	0.75	13.12	0.66	6.41	0.27	2.11	0.12
2	1.91	0.26	18.43	0.82	5.85	0.57	8.03	0.43	4.74	0.27	2.50	0.10
3	2.22	0.15	14.89	0.40	4.91	0.47	5.12	0.16	3.74	0.25	3.28	0.09
4	3.30	0.26	14.22	0.53	4.66	0.51	4.73	0.14	3.43	0.16	3.64	0.11
5	3.73	0.22	14.11	0.38	4.84	0.40	4.34	0.09	3.50	0.19	3.69	0.10
6	3.95	0.25	14.41	0.41	5.43	0.39	4.20	0.07	3.72	0.14	3.77	0.14
7	4.25	0.21	15.09	0.44	5.90	0.48	4.16	0.14	4.02	0.35	3.67	0.25
8	4.45	0.25	15.01	0.49	5.91	0.51	4.05	0.14	3.99	0.29	3.80	0.22
9	4.57	0.26	14.82	0.27	5.49	0.32	3.97	0.09	3.82	0.16	3.85	0.16
10	4.63	0.30	14.89	0.41	5.39	0.49	3.84	0.13	3.93	0.25	3.98	0.16
11	4.61	0.30	14.61	0.37	4.89	0.37	3.89	0.10	3.52	0.14	4.18	0.09
12	4.52	0.23	14.49	0.27	4.72	0.32	3.76	0.04	3.65	0.16	4.24	0.08
13	4.48	0.34	14.57	0.34	5.41	0.38	3.75	0.06	3.45	0.21	4.28	0.09
14	4.82	0.28	14.40	0.28	5.40	0.35	3.56	0.04	3.73	0.11	4.19	0.05

¹Feed composition is provided on a fresh matter basis. GE = gross energy.

corporated. Post hoc comparisons were performed using the Sidak correction.

The urine output during each postprandial period was quantitatively measured and corroborated by rapid weight loss recorded on the scales (Vorndran et al., 2025) enabling calculation of urine excretion relative to the initial postprandial weight. To estimate weight loss attributable to metabolic activity and evapotranspiration, the weight of excreted urine was added to the final weight measurement of each postprandial period. A linear regression was then performed between the initial postprandial weight and the adjusted final weight (i.e., the prefeeding weight of the subsequent postprandial period + urine). The slope of this regression quantified the rate of weight loss attributable to metabolic and evaporative processes (**MetEvap**), which is shown in Figure 1. The total weight lost during a postprandial period was derived from the same dataset to ensure comparability with MetEvap estimates. As previously stated for the hourly weight loss intervals, data from 6 out of 266 postprandial periods was also excluded from modeling due to the absence of weight loss. This occurred when urine excretion was high and milk intake was low during the same interval, as the addition of urine to the final weight measurement (for MetEvap estimation) exceeded weight loss. For each calf, 14 slope values (one per feeding) were derived and analyzed using a similar GLMM, with fixed effects for sex, feeding number, and period (day, night), and a random intercept for calf ID to account for individual variability. The interaction between feeding number and day and night period was also included. Again, a first order autoregressive correlation structure was incorporated to account for autocorrelation of repeated measurements

in calves over time. The variable day and night period was incorporated to account for differences in measurement resolution, as metabolic and total weight loss were calculated over the entire postprandial period rather than at hourly intervals as in the postprandial weight loss model. Both models were assessed for fit using the DHARMA package in R (<https://cran.r-project.org/web/packages/DHARMA/vignettes/DHARMA.html>). Diagnostic checks included the Kolmogorov-Smirnov test, dispersion test, outlier test, and inspection of quantile residual plots. All statistical analyses were conducted at a significance level of 0.05.

RESULTS

Accuracy and Suitability of the Scales

All 4 scales proved suitable for repeated measurements, consistently operating within the calibrated range. Deviations between repeated measurements remained within the manufacturer's specified sensitivity of ± 0.1 kg, confirming the reliability and precision of the equipment. Figure 2 provides a graphical representation of the calibration results, illustrating the stability and accuracy of the scales over time.

Weight Development

All calves consumed their dams' milk (colostrum, transition milk, and whole milk) throughout the first week of life at an average intake of 3.86 ± 0.97 kg per feeding. The results for feed consumption, composition, and performance are shown in Tables 1 and 2. The ADG within the first week of life was 1.31 ± 0.28 kg, notably ADG

Table 2. Performance data of 19 Brown Swiss calves fed twice daily at unlimited milk amount during the first week of life

Parameter	Male calves		Female calves	
	Mean	SEM	Mean	SEM
Birth weight, kg	46.34	1.60	42.29	1.94
End weight, kg	55.54	1.83	51.50	2.26
Milk intake, kg	54.84	3.41	52.16	2.71
DMI, kg	8.36	0.52	8.35	0.43
Feces excretion, kg DM ¹	0.14	0.05	0.12	0.04
Urine excretion, kg	28.92	2.25	24.76	2.24
Urinary DM excretion, kg	0.50	0.04	0.49	0.04

¹Due to methodological reasons, feces fresh matter could not be measured.

was measured using the classical approach of weighing at birth and d 7 on a separate scale and was not derived from the frequent scale data. Figure 3 presents a graphical illustration of the relative weight development from twice daily fed calves ($n = 19$) during their first week of life with unrestricted milk allowance per feeding. The fitted linear functions for each calf, normalized to their initial BW, are summarized in Table 3. The scale data exhibited a sawtooth-like pattern, with calf weight increasing abruptly after each feeding event and then decreasing continuously throughout the postprandial period (Figure 3). These variations are attributed to milk intake, metabolism of the feedstuff, urination, and evapotranspiration. The maximum BW during a postprandial period was always observed immediately after a feeding event, and the minimum immediately before the next feeding event.

Postprandial Weight Loss

The weight loss of all calves during all postprandial periods (14 periods during the first week) lay between 3.5% and 7.4%. Statistical analysis revealed that postprandial weight loss was significantly influenced by the feeding event ($P < 0.001$) and the duration of the postprandial period ($P = 0.026$). An influence of sex of the calf was absent; however, a trend toward an interaction between feeding event and duration of the postprandial period was noted ($P = 0.099$). Analysis of the feeding event effect revealed that weight loss during feedings 1 to 3 differed significantly from feedings 4 to 14 (Table 4). The mean rate of weight loss was low immediately after birth with approximately 0.346% BW/h. It increased progressively through the early feeding events and stabilized after the third feeding at 0.607% BW/h. However, when examining the postprandial rate of weight loss on an hourly basis over all feeding events (Figure 4A), a clear polynomial trend emerged: The highest weight loss occurred immediately after a feeding event, followed by a continuous decline in weight as time since the feed-

ing event increased. Significant effects of feeding event ($P < 0.001$), hours postprandial ($P < 0.001$), and their interaction ($P < 0.001$) were identified. A calf sex effect was absent. Feedings 1 to 3 and 4 to 14 were analyzed separately, based on the statistical difference mentioned earlier revealing group differences also in hourly weight loss until the sixth feeding event after birth (Figure 4B). Notably, the rate of weight loss failed to differ thereafter except for the 14th feeding event. The polynomial functions that best described weight loss per hour for both feeding groups up to 10 h after feeding were as follows: Feedings 1 to 3:

$$x = \text{hours after feeding (1–10),}$$

$$y = 0.904504 - 0.365716x + 0.073074x^2 - 0.005341x^3 + 0.000107x^4,$$

$$R^2 = 0.994;$$

Feedings 4 to 14:

$$x = \text{hours after feeding (1–10),}$$

$$y = 1.495098 - 0.236579x + 0.000288x^2 - 0.000111x^4,$$

$$R^2 = 0.999.$$

Metabolization and Evapotranspiration Losses

To isolate the effects of metabolization and evapotranspiration from total weight loss, the urine excretion during each particular postprandial period was added to the final data point of each observation period (Figure 1). The corresponding rates are presented in Table 4. Feeding event ($P < 0.001$) and calf sex ($P = 0.023$) exhibited significant effects on MetEvap. Male animals showed lower MetEvap losses than female animals with 86% relative to the rate of female animals. MetEvap rates at feedings 2 and 3 were lowest and differed from feeding 1 as well as from feedings 4, 7, and 9 to 14. In contrast, differences were absent between feeding 1 and feedings 4 to 14 with mean hourly MetEvap losses ranging from 0.152% to 0.304%. Expressed as a percentage of total weight loss, MetEvap losses accounted for 79% of weight loss after colostrum feeding (first feeding after birth), and between 33% and 49% during all following feeding events.

Correlations

Spearman rank correlation coefficients (r_{spearman}) of total BW loss and MetEvap losses to the composition

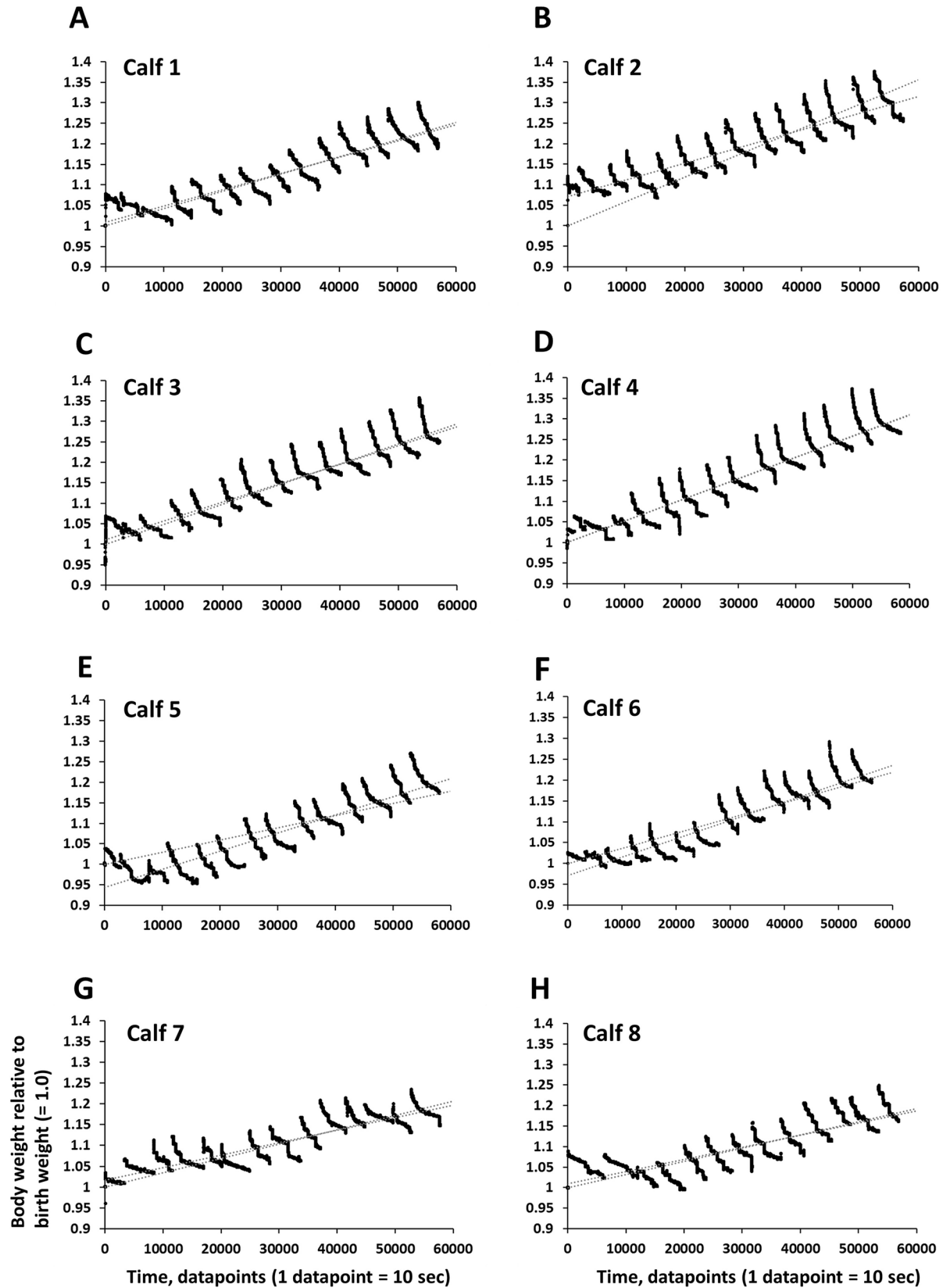


Figure 3. Relative weight development of neonatal Brown Swiss calves fed their dams milk twice daily in unrestricted amounts during the first week of life on a 10-s basis (1 datapoint = 10 s).

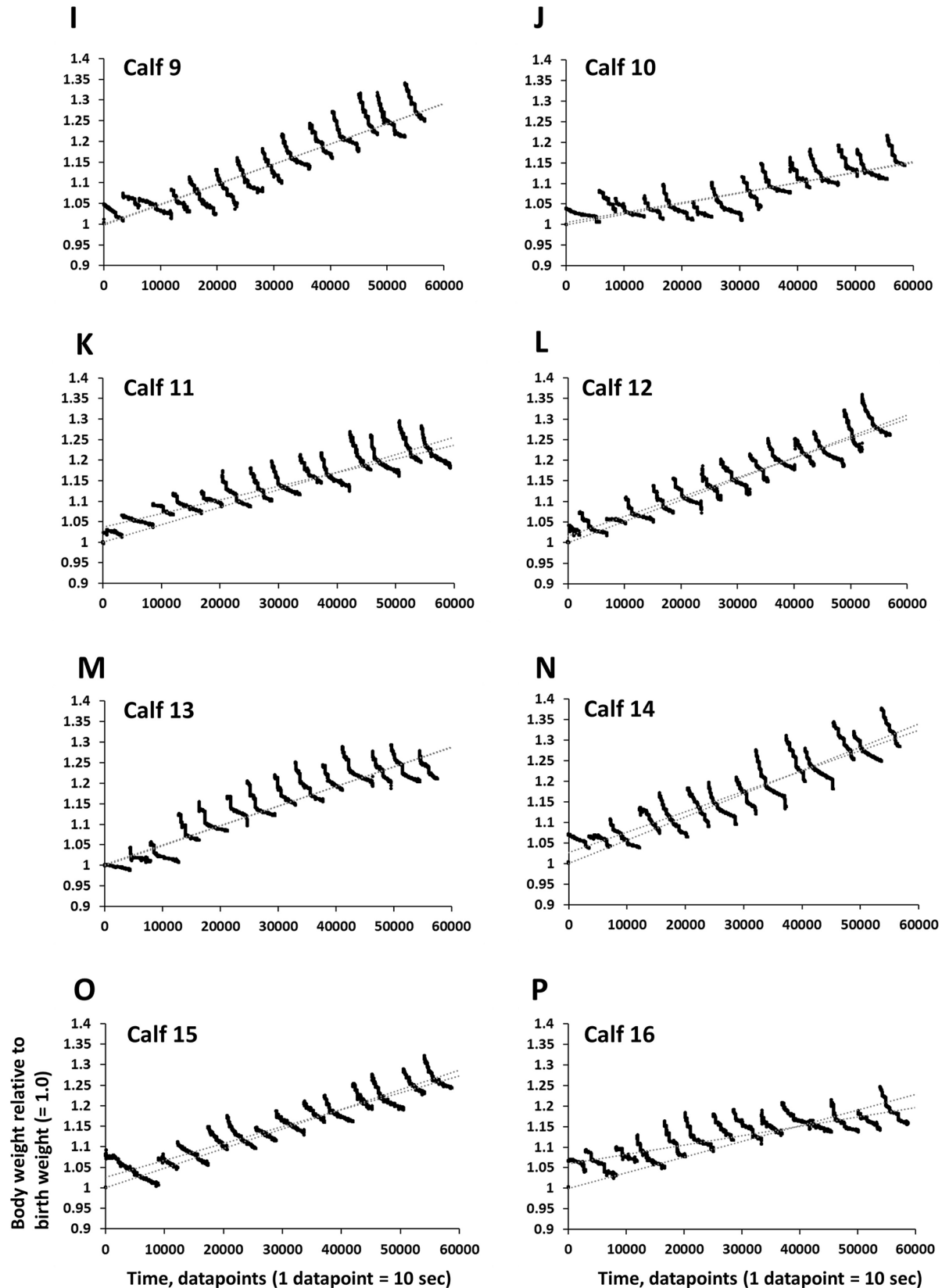


Figure 3 (Continued). Relative weight development of neonatal Brown Swiss calves fed their dams milk twice daily in unrestricted amounts during the first week of life on a 10-s basis (1 datapoint = 10 s).

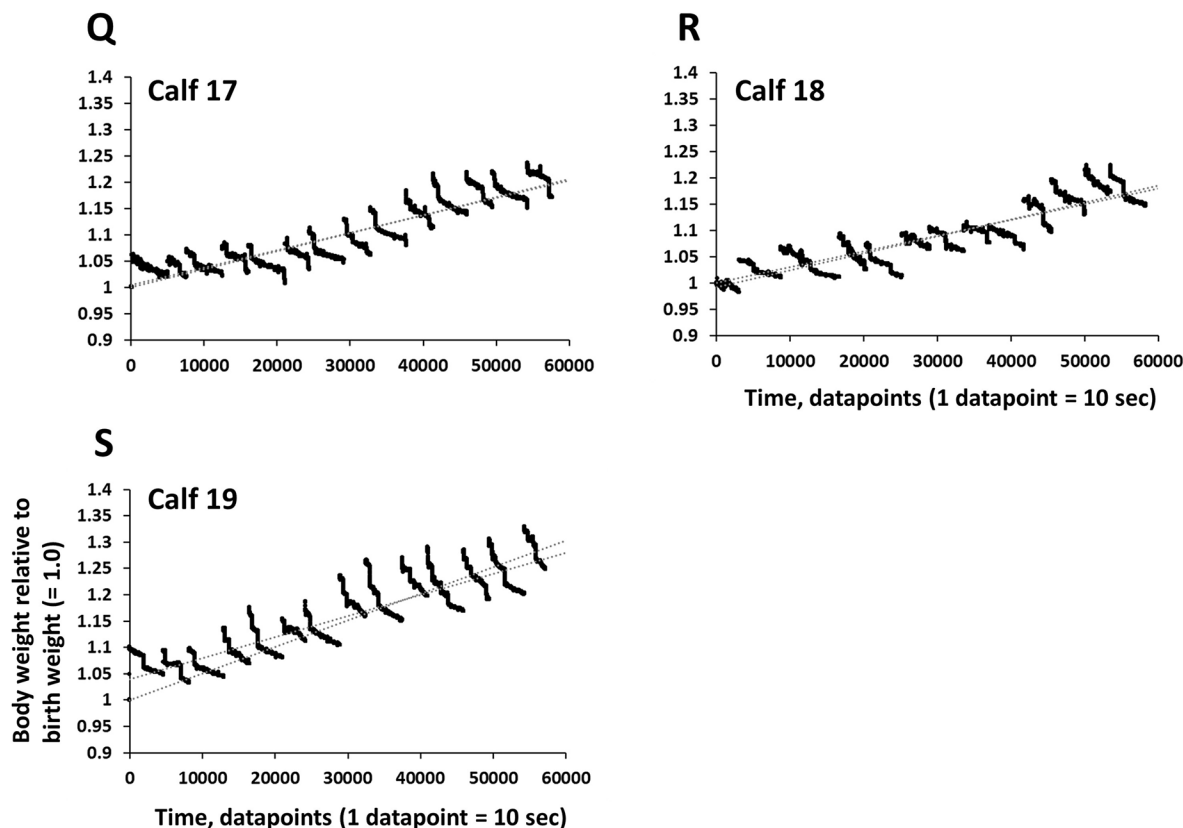


Figure 3 (Continued). Relative weight development of neonatal Brown Swiss calves fed their dams milk twice daily in unrestricted amounts during the first week of life on a 10-s basis (1 datapoint = 10 s).

of feedstuff and urine during the postprandial period are shown in Table 5. The highest correlation coefficients of weight loss and MetEvap losses were identified for urine excretion, followed by milk intake, which are themselves highly correlated ($r_{\text{spearman}} = 0.804$, $P < 0.001$). Correlation coefficients of feed and urine constituents to MetEvap were smaller than to total BW loss.

DISCUSSION

The presented models can be useful for interpreting short-term weight dynamics in neonatal calves under controlled feeding conditions. Specifically, they offer insights into the time-dependent physiological responses to milk intake, such as urination and MetEvap, and help characterize normal growth patterns and weight deviations during the first week of life. This is particularly relevant for assessing ADG at different time points and understanding the magnitude of postprandial weight loss as a physiological response to milk ingestion. Importantly, these models provide a first basis for visualizing and quantifying such dynamic processes in early calf development. Although they are tailored to a twice-daily unlimited dam's milk feeding regimen, the underlying

principles may inform feeding management decisions and monitoring strategies across different systems, provided their limitations are acknowledged. Deviations from this protocol, however, such as restricted or ad libitum feeding, are likely to result in different growth dynamics and may not align with the trends illustrated in the results. Furthermore, the applicability of our findings to other cattle breeds should be made with consideration of breed-specific thermoregulatory characteristics and physiological differences, such as variation in birth weight, coat thickness influencing the thermoneutral zone, and differences in the ratio of surface area to body mass affecting heat dissipation. The characteristic sawtooth pattern in weight data is, however, expected for all breeds and across all feeding regimens: restrictive, ad libitum, and mother bound. This pattern arises from the physiological response to milk intake. Given the strong positive correlation between the amount of milk intake and postprandial weight loss, it appears logical that feed restriction should reduce the magnitude of postprandial weight loss. Moreover, more frequent feeding events are expected to increase the number of postprandial periods, thereby raising the frequency of fluctuations in the weight curve, which would likely appear in ad libitum

Table 3. Linear regression coefficients describing the relationship between relative BW change (y) and 10-s interval measurements (x) during the first week of life in 19 Brown Swiss calves fed unrestricted dam's milk twice daily¹

Calf number	Linear regression with estimated t			Linear regression with t set to birthweight			Number of measurements
	m	t _{est}	R ²	m	t _{bw}	R ²	x
1	0.00000396	1.00907426	0.825	0.00000420	1.00	0.821	56.966
2	0.00000407	1.07139079	0.812	0.00000594	1.00	0.585	57.390
3	0.00000461	1.01107987	0.860	0.00000490	1.00	0.855	57.040
4	0.00000517	0.99971218	0.889	0.00000516	1.00	0.889	58.477
5	0.00000442	0.94360715	0.827	0.00000296	1.00	0.707	58.113
6	0.00000439	0.97158855	0.862	0.00000363	1.00	0.828	56.193
7	0.00000301	1.01663351	0.807	0.00000344	1.00	0.784	57.794
8	0.00000296	1.00987532	0.689	0.00000322	1.00	0.681	56.946
9	0.00000489	0.99805892	0.860	0.00000483	1.00	0.860	56.701
10	0.00000243	1.00471541	0.703	0.00000255	1.00	0.701	58.591
11	0.00000334	1.03641691	0.840	0.00000426	1.00	0.755	59.419
12	0.00000473	1.01583370	0.902	0.00000515	1.00	0.893	56.981
13	0.00000474	1.00245524	0.871	0.00000480	1.00	0.871	57.582
14	0.00000498	1.02603404	0.855	0.00000566	1.00	0.833	56.806
15	0.00000412	1.02558774	0.891	0.00000478	1.00	0.862	58.783
16	0.00000228	1.05973041	0.718	0.00000380	1.00	0.290	58.820
17	0.00000331	1.00464276	0.834	0.00000343	1.00	0.832	57.633
18	0.00000322	0.99173299	0.819	0.00000301	1.00	0.814	58.267
19	0.00000403	1.03894051	0.817	0.00000505	1.00	0.747	57.169

¹m is the slope of the regression reported as % BW change relative to birth weight/10 s. t_{est} describes the intercept that is estimated by the linear regression. t_{bw} describes the linear regression in which measured birth weight was used as an intercept.

fed or mother-bound calves that engage in more suckling bouts throughout the day (9 times/d during the first 3 wk of life) compared with restrictive feeding systems (Whalin et al., 2021). Therefore, the magnitude of postprandial weight loss and the frequency of postprandial periods can be influenced by feeding management, whereas the linear trend of weight gain should remain consistent across feeding systems and depend primarily on the quantity and nutritional composition of the feedstuff provided to calves during the first week of life and thereafter. The reported milk intake of the calves in our study, 3.86 ± 0.97 kg per feeding and 7.72 kg/d with twice daily feeding, compared well with reported milk intakes under ad libitum feeding conditions by Jasper and Weary (2002). They reported daily milk consumption during the first week of life ranging from 5 to 9 kg, with 7 to 9 kg from the third day onward and an average intake of 8.8 kg/d during the preweaning period from d 0 to 36. Kürn et al. (2016) also reported daily milk consumption under ad libitum feeding conditions at 6.90 ± 3.20 L/d during the first week of life. The ADG achieved in this study was very high and more comparable to growth rates reported under ad libitum feeding conditions rather than restricted feeding at 10% BW (0.8–1.02 kg/d; Hammon et al., 2002; Jasper and Weary, 2002; Kürn et al., 2016). Notably, the calves herein received their dam's milk, including transition milk, which was shown in a previous study by the authors to have a higher gross energy content compared

with milk replacer and whole milk, due to elevated fat levels (Vorndran and Steinhoff-Wagner, 2025). This likely explains an elevated growth performance observed during the first week of life compared with the mentioned ad libitum studies in which milk replacer or a whole milk pool was used instead of the transition milk.

The feed components DM and CP in DM were negatively correlated with postprandial weight loss, whereas others, such as crude ash in DM or lactose in DM were positively correlated. Fat content in DM failed to correlate to postprandial weight loss. However, the strongest correlation was observed between total milk intake and urine excretion. Therefore, attributing weight loss effects to individual constituents of milk or urine should be approached with caution, as the compositions of these substances change dynamically during the first week of life. Notably, milk composition shifts significantly during this period from colostrum and transition milk to whole milk resulting in a decrease in DM and nutrient concentrations (Vorndran and Steinhoff-Wagner, 2025). Interesting observations were made during the first 48 h after birth, where some calves consumed varying amounts of milk voluntarily but did not show significant weight gain. Instead, their BW remained relatively constant over the first 3 to 4 consecutive feedings and increased thereafter. Good examples of this phenomenon include calves 2, 6, 8, 10, and 18. During this period, calves were transitioned from bottle feeding to nipple bucket feeding, with

Table 4. Results for BW loss per hour after feeding, losses due to metabolization and evapotranspiration, and their proportions of total weight loss for 14 feedings of neonatal calves during the first week of life

Feeding	Weight loss, % BW/h				MetEvap loss, % BW/h				Urine, % of weight loss/h	MetEvap, % of weight loss/h
	Mean	SE	Lower 95% CI	Upper 95% CI	Mean	SE	Lower 95% CI	Upper 95% CI	Mean ¹	Mean
1	0.383 ^a	0.027	0.335	0.439	0.304 ^c	0.023	0.262	0.353	21	79
2	0.339 ^a	0.024	0.296	0.388	0.152 ^a	0.014	0.126	0.182	55	45
3	0.316 ^a	0.023	0.275	0.365	0.155 ^{ab}	0.015	0.129	0.187	51	49
4	0.521 ^b	0.031	0.463	0.586	0.228 ^c	0.018	0.195	0.266	56	44
5	0.657 ^c	0.037	0.588	0.734	0.220 ^{abc}	0.018	0.187	0.257	67	33
6	0.590 ^{bc}	0.035	0.526	0.661	0.221 ^{abc}	0.018	0.189	0.259	63	37
7	0.585 ^{bc}	0.034	0.523	0.655	0.233 ^c	0.018	0.201	0.271	60	40
8	0.585 ^{bc}	0.034	0.522	0.655	0.227 ^{bc}	0.018	0.195	0.264	61	39
9	0.608 ^{bc}	0.035	0.544	0.680	0.242 ^c	0.018	0.208	0.280	60	40
10	0.597 ^{bc}	0.034	0.534	0.668	0.265 ^c	0.019	0.230	0.306	56	44
11	0.646 ^c	0.036	0.579	0.721	0.258 ^c	0.019	0.223	0.299	60	40
12	0.621 ^{bc}	0.035	0.555	0.694	0.284 ^c	0.020	0.247	0.327	54	46
13	0.631 ^{bc}	0.036	0.565	0.705	0.262 ^c	0.020	0.226	0.303	58	42
14	0.637 ^{bc}	0.036	0.570	0.712	0.251 ^c	0.019	0.216	0.291	61	39
<i>P</i> -value										
Feeding	<0.001				<0.001					
Sex	0.526				0.023					
Morning or evening	0.419				0.226					
Feeding × morning or evening	0.362				0.374					

^{a-c}Superscripts indicate significant differences ($P < 0.05$) based on the transformed data at an α level of 5%. Statistical comparisons were performed on log-transformed data and back transformed.

¹Calculated by difference from % MetEvap losses. MetEvap = metabolization and evapotranspiration. Data are presented as LSM $\pm$ SE.

varying degrees of assistance necessary from the caretakers. Some calves struggled for example with getting up, keeping balance, tongue placement, head extension, or finding the nipple bucket. In contrast, calves 7, 11, and 12 voluntarily ingested large amounts of milk from the bottle and adapted quickly to the bucket, which gave them an early weight advantage. This shows, that learning to drink from the nipple bucket without assistance early is crucial to secure weight gain in the first days after birth. In a study by Mandel et al. (2017), the average time for Holstein calves to drink without assistance from a bucket (no nipple bucket) was 3 d; however, wide variation was described, with some calves requiring 5 h to drink on their own and others >10 d. In this study, no calf had to be assisted for longer than 4 d, with most calves learning to successfully interact with the nipple bucket after 2 assisted feedings.

To the best of our knowledge, postprandial weight loss in neonatal calves has not previously been documented in comparable detail, which currently precludes direct comparison with data from preweaning animals. In the absence of such reference data, it is necessary to consider the physiological characteristics specific to the neonatal period that may influence early weight dynamics in contrast to adult animals, and these factors should be taken into account when interpreting our results. Although neonatal calves exhibit functional thermoregulatory mechanisms at birth, approximately 2% of their body

mass consists of brown adipose tissue, which facilitates nonshivering thermogenesis (Roland et al., 2016). This brown adipose tissue is most abundant in the perinatal period and is essential in preventing hypothermia for calves exposed to cold conditions (Symonds et al., 2015). Furthermore, the ingestion of colostrum contributes to endogenous heat production, thereby supporting thermal homeostasis during early postnatal life, which is necessary, as calves cannot rely on heat production from rumen fermentation, as adult animals can (Roland et al., 2016). Also, distinct differences in the renal function of neonates to adult animals have to be acknowledged. Compared with adult animals, neonates have a limited ability to concentrate urine, although their capacity to dilute urine appears to be comparable (Liu et al., 2001; Quigley et al., 2004). As previously published by Vorndran et al. (2025) the urine of neonatal calves undergoes notable compositional changes during the first week of life. Immediately after birth, urine contains the highest concentrations of DM and various metabolites, which progressively decline when calves are provided with a high plane of nutrition and sufficient fluid intake. Also, the concentrations of nucleic acids were most abundant in the earliest urine samples, indicating a period of elevated nucleic acid turnover immediately after birth or potentially even during late gestation (Vorndran et al., 2025). Following colostrum ingestion, neonatal calves exhibit transient proteinuria in urine lasting up to 30

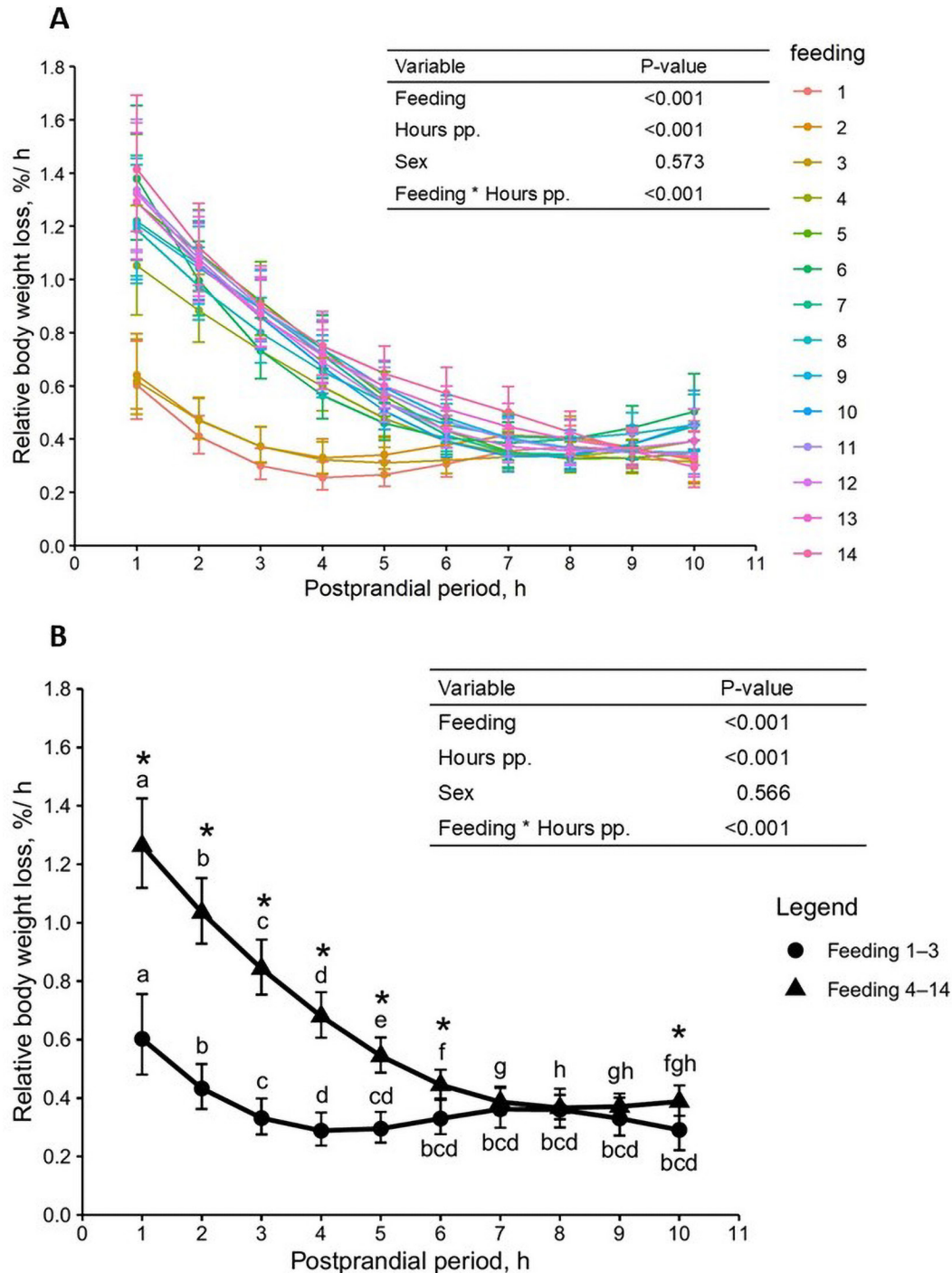


Figure 4. Interaction effect of the feeding event and the hours postprandial on % BW loss per hour of neonatal Brown Swiss calves fed at voluntary milk intake ($n = 19$, A) and the relative hourly BW loss (%) following voluntary milk intake in neonatal Brown Swiss calves ($n = 19$), comparing feedings 1 to 3 (immediate postnatal period), and feedings 4 to 14 (subsequent feedings during the first week of life, B). Lowercase letters indicate significant differences within the grouped feedings. * indicates significant differences between the grouped feedings.

to 48.5 h (Pierce, 1959; Vorndran et al., 2025), which has been described by Skrzypczak et al. (2021) as an adaptive renal mechanism to eliminate excess proteins during the period of intestinal permeability that enables

passive transfer of immunoglobulins (Weaver et al., 2000). Although proteinuria is a normal phenomenon in neonatal calves up to 2 d of age, it is an indicator of urinary tract diseases in adult ruminants (Divers, 2018).

Table 5. Spearman rank correlations of postprandial weight loss to milk constituents, milk intake, and urine excretion

Item	Weight loss			MetEvap ¹ loss		
	Correlation coefficient	P-value	Sample size	Correlation coefficient	P-value	Sample size
Milk parameters						
Milk intake	0.729	<0.001	263	0.422	<0.001	263
Brix	-0.442	<0.001	264	-0.177	0.004	264
Dry matter	-0.537	<0.001	265	-0.106	0.086	265
Fat in DM	-0.096	0.129	251	0.027	0.675	251
CP in DM	-0.308	<0.001	251	-0.213	<0.001	251
Lactose in DM	0.741	<0.001	139	-0.324	<0.001	139
Crude ash in DM	0.482	<0.001	140	0.005	0.949	140
DMI	0.522	<0.001	248	0.397	<0.001	248
Fat intake	0.353	<0.001	248	0.328	<0.001	248
CP intake	0.343	<0.001	248	0.300	<0.001	248
Lactose intake	0.765	<0.001	138	-0.367	<0.001	138
Urine parameters						
Urine excretion	0.811	<0.001	256	0.235	<0.001	256
DM	-0.661	<0.001	250	-0.365	<0.001	250
pH value	0.436	<0.001	253	0.298	<0.001	253
Crude ash in DM	0.528	<0.001	134	0.381	<0.001	134
Nitrogen in DM	-0.133	0.039	241	-0.138	0.032	241
Urea in DM	0.199	0.029	120	0.152	0.098	120

¹MetEvap = metabolism and evapotranspiration.

This illustrates how physiological processes in neonatal calves are closely linked to developmental adaptations that are influenced by colostrum intake, which is a fundamental aspect of neonatal calf nutrition, not only due to its role in passive immunization, but also because of its influence on thermoregulation and the development of the gastrointestinal tract (Weaver et al., 2000; Blum and Baumrucker, 2002; Roland et al., 2016). In addition to immunoglobulins, colostrum contains a wide range of bioactive compounds including hormones, growth factors, and oligosaccharides, which contribute to both the structural and functional maturation of the gastrointestinal tract (GIT; Campana and Baumrucker, 1995; Blum and Baumrucker, 2002; Huang et al., 2025). Colostrum feeding has been shown to increase villus height and crypt depth in the GIT of neonatal calves when compared with calves fed an identical formula diet lacking bioactive compounds (Steinhoff-Wagner et al., 2014). As a result, the absorptive surface area in the GIT is expanded, which has been associated with improved glucose absorption and metabolism (Hammon and Blum, 1997; Steinhoff-Wagner et al., 2011, 2014). All the aforementioned mechanisms occur during the immediate neonatal phase, during which this study investigated the growth dynamics and postprandial weight loss. Although the developmental stage differs, the shared mechanisms of weight loss by GIT emptying, urination, shifts in water balance and respiration (Self and Gay, 1972; Phillips et al., 1991; Coffey et al., 2001) remain comparable between neonates and adults.

In adult animals, shrink is a well-documented phenomenon resulting from feed and water restriction for example during the transport to fattening or rearing facilities (Coffey et al., 2001). Teke (2013) documented that adult bulls transported to a slaughter facility over a 30-h period, including a 2-h rest interval, exhibited BW losses ranging from 1.7% to 8.39%, with a mean shrinkage of 5.5%. Beef cattle held in a dry lot are expected to experience shrink losses of 3.3% to 6.2% of BW over an 8 to 16 h period, whereas cattle transported in a moving truck may lose between 5.5% and 7.9% of BW (Mullenix et al., 2018). Notably, the observed weight loss in neonatal calves in this study ranged from 3.5% to 7.4% of BW between feeding and 10 to 14 h postprandial, aligning closely with the shrinkage values reported for beef cattle standing in the dry lot although BW of the animals was very different.

Due to the high-frequency weight data, we were able to describe postprandial weight loss to be of polynomial nature. Also, Heitschmidt (1982) found that a curvilinear model described early weight loss in cows during the first 7 h in a dry lot more accurately, whereas both linear and allometric models performed similarly between 7 and 22 h after the last feeding, making linear regression adequate for most practical purposes of shrink calculation. The prediction equation reported by Heitschmidt (1982; $y = 2.43 + 0.35x$) yields a loss of 0.35% of BW every hour after an initial weight loss of 3.48% in 3 h, which translates to 1.16% of BW per hour immediately after feeding. The observed results from our study lay in the same

region for feeding events 4 to 14, when calves consumed higher amounts of feed (3.3–4.8 kg/feeding), with 1.26%, 1.04%, 0.84%, 0.68% BW loss during postprandial hours 1 to 4. Also, Coffey et al. (2001) stated that shrinkage for beef cattle could be assumed with 1% BW loss/h for the first 3 to 4 h of feed and water withdrawal, and after 10 h, the shrinkage decreased to 0.1%/h. In a study by Fernandez et al. (1996) on 20-wk-old veal calves, the shrink during the first hour of transport was 0.77% BW/h and for the next 10 h 0.29% of BW/h. Analogue to our results described for postprandial weight loss data, the BW loss is higher closer to the last feeding, although with 1.26% weight loss during the first hour postprandial, our results appear higher compared with the data documented by Fernandez et al. (1996). However, the quality of feedstuff (digestibility and passage rate) and feed amounts play a role in the magnitude of weight loss (Cravey et al., 1991). Twice daily unrestricted feeding of calves might have led to higher weight loss as compared with restrictive feeding before the fasting period. Furthermore, the complete absence of solid feed intake in our study makes comparability to older calves that ingest solid feed and have a functional forestomach system challenging. The higher intake of solid feed compared with milk, which retains moisture during digestion, may reduce weight loss for a prolonged period of time after feeding. Contrary, milk is expected to increase weight loss in the initial hours of the postprandial period while potentially reducing weight loss later due to its high water content. This thought is supported by a study from Cravey et al. (1991), which found that cattle previously fed on wheat pasture experienced greater shrink under feed restriction during transport than those fed hay. The reduced shrinkage in hay-fed cattle was attributed to the hay's high DM content, which resulted in a slower passage rate of the feedstuff. Also, a study by Abe et al. (1999) conducted on 1-wk-old calves fed various combinations of starter, hay, and milk replacer found that urine amounts were lower in calves receiving dry feed and milk replacer compared with those fed only milk replacer. However, when water was made available alongside dry feed, the urine amount was comparable to that of calves fed only milk replacer. This again shows that the inclusion of dry feed, such as hay, influences fluid balance and excretion patterns, potentially contributing to reduced shrinkage. A prolonged GIT fill can therefore be regarded as beneficial in minimizing shrinkage during periods of feed restriction. Supporting this, Self and Gay (1972) reported that approximately 50% of the weight loss observed during the transport of yearling steers was due to GIT emptying, while carcass loss accounted for 30%. Notably, the weight lost due to GIT depletion is rapidly regained once animals are reintroduced to feed (Cravey et al., 1991). Transportation is recognized as a significant stressor af-

flicting animal behavior (Swanson and Morrow-Tesch, 2001), which can temporarily increase fecal and urinary excretion as well as the mobilization of nutrient reserves in calves compared with not transported animals (Phillips et al., 1991). Phillips et al. (1991) found that feeder calves transported for 48 h exhibited significantly greater urine excretion but no changes in fecal excretion were observed compared with calves that were fasted for 48 h at a barn. Specifically, urine accounted for 34.05% of total weight loss in the transported group, versus 23.90% in calves subjected only to fasting in a barn. This increased urine output explained approximately 80% of the additional weight loss observed due to transport. Age also influenced the excretory losses with younger calves (9 mo old) excreting more urine than older calves (16 mo old) and urine contributing between 15.20% and 18.96% to total weight loss. Fecal losses accounted for 34.27% (barn) and 37.02% (transport), and combined, urine and feces accounted for 60.92% (barn) to 68.32% (transport) of the total weight loss during the fasting period (Phillips et al., 1991). The reported values for urine excretion appear substantially lower than those measured in neonatal calves in this study, where urine accounted for 51% to 67% of total weight loss. This discrepancy may again be explained by the exclusive milk feeding in preweaning neonatal calves, in contrast to weaned calves that consume significant amounts of dry feed, which retains moisture. Although direct comparisons to beef cattle and older calves present certain challenges, we would like to highlight, that the results of our study on postprandial weight loss in neonatal calves, normalized for body weight, compared surprisingly well and were remarkably consistent. Similarly, Phillips et al. (1991) described in the previously explained study that older steers lost more weight and excreted more feces and urine (kg) compared with younger animals, but calculated as a percentage of BW, the excretions were similar between the groups.

The NASEM (2021) specifies the maintenance energy requirement of a 45-kg calf from birth to the third week of life under thermoneutral conditions at 1,859 kcal ME/d. To evaluate our MetEvap calculations, the energy requirement was converted into necessary milk consumption as a percentage of BW using the reported gross energy (GE) values for the consumed milk. During d 1 to 3 of lactation, the milk contained 4.26 kJ GE/g fresh matter (FM), and during d 4 to 7, 3.76 kJ GE/g FM. To convert GE into ME, a digestibility of 95% and a metabolizability of 97% for whole milk was assumed based on the values given by the NASEM (2021). The corresponding ME values were 3.96 and 3.50 kJ ME/g FM, respectively. To match the energy requirements for maintenance a daily milk intake of 1.96 kg (d 1–3) and 2.22 kg (d 4–7) would therefore have been necessary. Expressed as a percentage of BW of a 45-kg calf, this corresponds to 4.26% to 4.91% BW/d,

or a mean weight loss of 0.178% to 0.204% BW due to metabolization of the feedstuff per hour (assuming a 24-h period). In comparison, our calculated MetEvap losses exceeded these estimates by approximately 20% to 30%. This overestimation was likely due to evaporative losses from the animals and straw bedding, which represents a limitation of the study. Under the described conditions, the scale-based measurements could not reliably separate metabolic losses from evapotranspiration. Respiration chambers would be required for accurate quantification. Another limitation was the inability to directly measure feces excretion, as feces remained in the bedding until the end of the trial. To account for this, the overestimation caused by feces and retained moisture was quantified after calves left the scales and interpolated across the dataset. This correction preserved the integrity of weight dynamics throughout the study and adjusted final BW values accordingly, ensuring that linear regression analyses were not affected by this systematic error. Removal and drying of bedding material at the end of the experiment enabled determination of DM excretion by feces, which averaged 130 g/calf by the end of the first week of life. Based on an estimated milk intake of 53 kg with 14% DM and high digestibility (95%–99%), the expected undigested and excreted DM would range between 0.07 and 0.37 kg, excluding microbial metabolism and gaseous losses, supporting our measurement. This amount likely had a small influence on MetEvap and weight loss observations compared with urine, which was quantitatively collected and totaled approximately 26 kg (liquid) per calf. Urine primarily drained quickly over the straw, as the slatted floor beneath facilitated rapid removal, resulting in minimal urine retention by the thin straw layer. Assuming a DM concentration of 15% to 20% of feces (Renaud et al., 2020), the total fecal FM accumulated on the scales would have been less than 1 kg during the first week of life. Thus, any potential effect on postprandial weight loss measurements during the 14 observation periods would have been smaller than the scale's sensitivity (± 0.1 kg). The reported postprandial weight loss results are therefore considered robust; however, it should be noted that, if anything, weight loss may have been slightly underestimated due to the retention of feces on the scale during the observation periods. An overestimation of weight loss can therefore be excluded because the scales only detect a weight loss if something leaves the scale, such as urine. Based on the listed limitations, the results generated with this novel experimental approach should be considered exploratory.

The first postprandial period following colostrum intake differed from subsequent intervals, likely due to a delay between feed ingestion and the onset of urine excretion into the collection canister. During this initial phase, calves showed less frequent urination and high

fluid retention (Vorndran et al., 2025), resulting in lower weight loss and a higher proportion of MetEvap relative to total weight loss, with urine accounting for only 21% of weight loss during the postprandial period. This elevated MetEvap contribution may also be caused by the high DM content of colostrum and the lower milk intake immediately after birth. Additionally, amniotic fluid or saliva retained in the calves' coat after birth may have contributed to increased evapotranspiration during this initial period as a result of coat drying. An increased respiratory rate after birth (Vannucchi et al., 2018) may also have increased evaporative losses, however the lungs volume increases over time (Castleman and Lay, 1990).

A significant sex effect was observed for MetEvap, with males exhibiting lower values than females. This was attributed to females consuming less milk, resulting in reduced urine excretion compared with males. Because MetEvap was calculated based on the proportion of urine excretion relative to total weight loss, higher urine output lead to a smaller MetEvap contribution to overall weight loss. The increased contribution of urine to weight loss in later phases of the first week of life corresponded to increasing milk intake and supported the interpretation that early postprandial weight loss was more influenced by metabolization and evapotranspiration losses than by urination, although the specific contribution of each factor remained unknown.

CONCLUSIONS

During the first week of life, calf growth followed a linear trajectory. However, daily fluctuations in body weight resulted in a distinctive sawtooth-like growth pattern, which was influenced by feeding management. Postprandial weight loss correlated with milk intake and urine excretion, and these two variables were interdependent. The magnitude of postprandial weight loss varied depending on the feeding event and the duration of the postprandial period. The dynamics of weight loss were best described by fourth-degree polynomial models, which compared well with literature data on shrinkage in older animals after feed deprivation, with the highest weight loss observed in the hours closest to the last feeding and a decreasing magnitude thereafter. The models presented offer valuable insight into early growth dynamics and support more precise interpretation of naturally occurring weight changes in neonatal calves, which may help improve the accuracy of weight measurements and growth assessments in future animal studies.

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Nonstandard abbreviations used: GE = gross energy; GIT = gastrointestinal tract; GLMM = generalized linear mixed model; FM = fresh matter; MetEvap = metabolization and evapotranspiration; r_{spearman} = Spearman rank correlation coefficient; t_{bw} = the linear regression in which measured birth weight was used as an intercept; t_{est} = the intercept that is estimated by the linear regression.

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ORCIDS

Anton Maria Vorndran, <https://orcid.org/0009-0000-7878-1589>

Heinz Bernhardt, <https://orcid.org/0000-0003-1080-2907>

Julia Steinhoff-Wagner <https://orcid.org/0000-0003-0180-9859>